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| **Fiscal Year:** | FY 2015 | **Task Last Updated:** | FY 01/15/2015 |
| **PI Name:** | Su, Ching-Hua Ph.D. | | |
| **Project Title:** | Crystal Growth of II-VI Compound Semiconductors | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | MATERIALS SCIENCE | | |
| **Program/Discipline--Element/Subdiscipline:** | MATERIALS SCIENCE--Materials science | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Organization Name:** | NASA Marshall Space Flight Center | | |
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| **City:** | Huntsville | **State:** | AL |
| **Zip Code:** | 35812 | **Congressional District:** | 5 |
| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | NOT AVAILABLE |
| **Start Date:** | 10/01/2006 | **End Date:** | 09/30/2016 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 1 | **Monitoring Center:** | NASA MSFC |
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| **Flight Program:** | ISS | | |
| **Flight Assignment:** | NOTE: Change in end date per P. Downey/MSFC (Ed., 9/28/2011) | | |
| **Key Personnel Changes/Previous PI:** | Feb. 2013 report: Co-Investigator: Dr. Narayanan Ramachandran, Jacobs ESSSA Group, ER42, NASA Marshall Space Flight Center. | | |
| **COI Name (Institution):** | Ramachandran, Narayanan ( Jacobs ESSSA Group )<br>Dudley, Michael ( Stony Brook University ) | | |
| **Grant/Contract No.:** | Internal Project | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | Two reasons have been put forward to account for the improved crystal quality when crystals are grown in a microgravity environment. The first is weight-related reductions in crystal strain and defects. These are thought to be caused by the weight of the crystals during processing at elevated temperatures and retained on cooling, particularly for materials with low yield strength. The second and more general reason is related to the reduction in density-gradient driven convection in the fluid during crystal growth in reduced gravity. The objectives and experiment description for the proposed flight experiment “Growth of Ternary Compound Semiconductors” by (1) vapor growth and (2) melt growth are described below.<br>1a) Vapor Growth<br><br>The physical vapor transport (PVT) crystal growth process consists of essentially three processes: sublimation of source material, transport of vapor species, and condensation of vapor species to form crystal. The latter two processes can be affected by the convection caused by gravitational accelerations on Earth. Reductions in such convection in low gravity is expected to yield a nearly diffusion-limited growth condition which results in more uniform growth rates (on the microscopic scale) and hence greater crystalline perfection and compositional homogeneity. This reduction of convective contamination in a reduced gravity environment will help to simplify the complexity of the coupled mass transport and growth kinetics problem. An improved comparison between the experimental results and theoretical simulations is also expected and will be beneficial to any growth process involving vapor mass transport or vapor-crystal interface growth kinetics.<br><br>The materials to be studied in this project are ZnSe and the related ternary compound semiconductors such as ZnSeTe, ZnSeS, and ZnCdSe. The scientific objectives of the growth of ZnSe crystals in reduced gravity using physical vapor transport (PVT) processes are:<br><br>• to establish the relative contributions of gravity-driven fluid flows to (i) the non-uniform incorporation of impurities and defects and (ii) the deviation from stoichiometry observed in the grown crystals as a result of buoyancy-driven convection and irregular fluid-flows caused by growth interface fluctuations and interface evolution.<br><br>• to assess self-induced strain effects developed during processing at elevated temperatures and retained on cooling caused by the weight of the crystals from x-ray topography and by measuring the density and distribution of various crystalline defects.<br><br>• to obtain high quality space-grown materials for various thermophysical and electrical properties measurements and as substrates for device fabrication and thus assess device performance as influenced by a substantial reduction in gravity-related effects.<br><br>1b) Melt Growth<br><br>The melt growth of homogeneous ternary compound semiconductors, such as cadmium zinc telluride (Cd1-xZnxTe, $0 < x < 1$) is difficult because the thermal and compositional gradients in the melt just ahead of the growth interface during directional solidification can cause substantial gravity-driven fluid flows that alter the relative distribution of the two (CdTe and ZnTe) binary components and introduce structural defects into the grown crystals. The scientific objectives of the growth of CdZnTe crystals in reduced gravity using directional solidification process are:<br><br>• to investigate quantitatively the influences of gravitationally dependent phenomena on the growth process and the incorporation of impurities and defects in the resulting crystals.<br><br>• to better understand and control the growth parameters in the production process on Earth so as to increase the yield of usable material from the grown ingot.<br><br>• to assess strain effects developed during processing at elevated temperatures and retained on cooling, caused by the weight of the crystals and, possibly, the contact with the growth container, from the structural characterization and by measuring the density and distribution of various crystalline defects.<br><br>• to obtain high quality space-grown materials for various electrical and optical property measurements and as materials for the x-ray and gamma-ray detector fabrication as well as substrates for infrared device application and thus assess device performance as influenced by a substantial reduction in gravity-related effects. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | • In crystal growth there is a need to understand the relation between processes in the fluid phase (both liquid and vapor), such as buoyancy driven convection and the incorporation of impurities, and defects in the resulting crystal<br>• Relation between fluid phase processes and the generation of defects in a grown crystal is an outstanding problem in materials growth<br><br>• Crystal quality greatly influences important electronic properties in materials • Ternary compound semiconductors are of vital national interest as sensors in x-ray telescopes and for homeland security, and as substrate materials for infrared sensors<br><br>• Studies in microgravity will be compared with modeling and will be used to optimize ground-based experiments |
| **Task Progress:** | 1. Several single crystals of cadmium zinc telluride grown by the modified Bridgman method were characterized by synchrotron X-ray topography at Brookhaven National Lab. Uniform distribution of dislocations and subgrain boundaries were observed. Higher distortion along edge could be attributed to lattice curvature or presence of high density of defects that turn them into a mosaic structure.<br>2. A New Technology Report, “A Method to Process High Efficiency Functional Gradient Thermoelectric Materials,” was submitted. Thermoelectric materials convert thermal energy, i.e. heat, into electric energy. The converting efficiency is usually rated by the values of Figure of Merit, zT, which varies as a function of temperature and shows a maximum at a certain temperature TM. Common thermoelectric materials have uniform composition and, when positioned in a thermal gradient, only a small portion of the material is operating in a narrow temperature range with high zT values. It is claimed, in this invention, that by carefully processing some compound semiconductors from the melts the resultant materials can possess thermoelectric properties such that TM within each small material layer varies monotonically |

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| | along the axial direction of the processed ingot. This unique property of the so-called Functional Gradient Thermoelectric Material will improve the total converting efficiency of the thermoelectric component because, as the ingot is operating in a thermal gradient, each layer of the material along its axis is performing close to the temperature of the highest efficiency. |
| **Bibliography Type:** | Description: (Last Updated: 02/18/2025) |
| **Articles in Peer-reviewed Journals** | Li C, Su C-H, Lehoczky SL, Scripa RN, Ban H, Lin B. "Thermophysical properties of HgTe and Hg0.9Cd0.1Te melts." Journal of Non-Crystalline Solids. 2014 May;391:54-60. http://dx.doi.org/10.1016/j.jnoncrysol.2014.03.012 , May-2014 |